

IWATSU

Digital Oscilloscopes

DS-5600A Series

100MHz to 500MHz

DS-5400A Series

100MHz, 200MHz

- Large memory of 5M points (when channels interleaved)
- Pass/Fail judgment waveform parameters increased to four
- Averaging performed a maximum of 65,536 times
- High-resolution mode function providing a vertical resolution equivalent to a maximum of 12 bits
- Supports the UART, I²C and SPI serial triggers
- Double calculations possible (differential calculus, integral calculus and FFT possible after addition, subtraction and multiplication)
- Supports transparent PNG image data



VIEWGO II

New Functions Providing Additional Power

ViewGo II

New Functions Providing Additional Power

(Functions supporting the DS-5600A series)

NEW

Long memory up to a maximum of 5M points (page 4)

Maximum memory length has been increased from 1M points to 5M points (when channels interleaved).

Enables long-term waveforms to be captured while maintaining high-speed sampling.

NEW

It is now possible to simultaneously set four waveform judgment parameters (page 4)

Waveform parameters used with the judgment function have been increased from one to four.

Performing detailed judgment of measurement signals is now possible.

NEW

Supports the new noise-reduction function (page 6)

■ Simple Moving Average

The Simple Moving Average (SMA) function has been added to the digital filters that can be set for each individual channel. This enables smoothing and reduces noise for the sampling points at the specified width.

■ Averaging Count Increased

The averaging count setting has been increased from 256 times to 65,536 times. This enables non-synchronized random noise signals to be effectively reduced from measured repetitive signals.

■ High Resolution

This reduces random noise and increases vertical resolution to a level equivalent to a maximum of 12 bits during waveform capturing.



DS-5654

NEW

Supports the UART, SPI and I²C serial triggers (page 6)

NEW

Other new functions

■ Waveform calculation function strengthened (double calculations) (Page 7)

This enables double calculations for performing FFT analysis on CH1 x CH2 multiplied waveforms.

■ Hard-copy image format strengthened (page 7)

Supports PNG transparency attributes to make waveforms transparent and overlay them when creating documents for use on a PC.

Refined Functions

DS-5600A Series

Waveform mask / Waveform Parameter Judgment, Logging Display

Establishes the Pass/Fail judgment for waveform mask and waveform parameter values and displays a list of the results (parameter/logging function). These results can be saved onto a USB memory.

DS-5600A Series | DS-5400A Series

Automatic Measurement for Waveform Parameters

26 types of waveform parameters for automatic measurement are now available. This also supports automatic measurement within ranges specified with the cursor function.

DS-5600A Series | DS-5400A Series

Parameter Measurement for All Cycles

The automatic measurement of waveforms for all loaded cycles (on a time axis) is provided. A vast amount of data can be loaded into the long memory, which enables the differentiation between discrepancies in maximum and minimum values.

DS-5600A Series

High-Speed Remote Waveform Transmission

The high-speed transmission of waveform data onto PCs has been continued on from the DS-5500 (A) series. This brings out the best performance on analyses using PCs.

* When the waveform display mode is set at OFF during LAN use.

Speed and ease of use condensed into a compact body

1 Touch-screen Display

Equipped with a 7.5 inch color liquid crystal display touch screen function to enable intuitive operations.

2 Touch-screen Area

Simple menu selection with one-touch operations.

3 AUTO SETUP Button

Automatically sets the voltage range, time range and trigger conditions in accordance with the received signals to enable waveforms to be observed and measured easily.

4 REPLAY Knob

Allows waveforms captured in the past to be displayed.
DS-5600: Max. 2,048 waveforms
DS-5400: Max. 1,024 waveforms

5 MEASURE (Auto-measure) Button

- Automatic waveform parameter measurement
- Waveform mask and waveform parameter judgment
- Measurement and analysis of parameter lists (waveform logs)*

* Only supported by the DS-5600A Series

6 CLEAR Sweep Button

Clears measurements and the calculation results. Items eligible for clearing: Replay waveforms, averaging process, persistence, auto-measure results (judgment and logging).

7 HORIZONTAL (ZOOM) Button/Knob

In addition to operating the input for each channel and the horizontal axis for waveform calculations, the ZOOM button enables the simple zoom-in of waveforms.

8 VERTICAL CH/MATH Button/Knob

The buttons and knobs are separately provided for each channel input, which provides smooth operations. The LIGHT TRACE button firmly switches the display ON and OFF.

9 POWER Switch

Provides high-speed start-up within seconds of the power being switched on.

10 USB Terminal (Supports Hi-Speed)

Enables hard copies, waveform data and panel setup data to be saved on a USB memory.

11 Channel Input, External Trigger Input

- Automatically detects probe attenuation rates.

* When probes fitted with lead-out pins are used.

- Equipped with a special external trigger input function.

12 Calibration Signal

Used to adjust probe phasing.
* Square-wave output: 1kHz, 3Vp-p

13 GPIB Interface

(DS-576: Factory-delivered option)
Supports IEEE488.2.

14 Expansion Terminal

The terminal to which the DS-579 probe power option and IE-1226 VGA Video Out option are connected.

15 USB and LAN* Interface

It is possible to connect this to a PC for remote control purposes. Connecting a printer that supports PictBridge to the USB terminal allows screen shots to be printed directly.
* Only supported by the DS-5600A Series

16 AUX I/O*

(DS-577/DS-578: Factory-delivered option)
Support CH Out and Trigger Out.
- CH1/CH2 output (DS-577), CH1/TRIG output (DS-578)
* Only supported by the DS-5600A Series



Rear of the DS-5654

Comparison Chart of DS-5600A Series and DS-5400A Series Functions

	DS-5600A Series	DS-5400A Series
Frequency Bands		
100MHz	✓	✓
200MHz	✓	✓
350MHz	✓	
500MHz	✓	
Maximum Memory Length		
During Channel interleaved	5M points	500k points
When All Channels are in Use (/CH)	2.5M points	500k points
Maximum Sampling Speed		
During Channel interleaved	2GS/s	2GS/s (DS-542x only)
When All Channels are in Use	1GS/s	1GS/s
Maximum Waveform Capture Time at the Maximum Sampling Speed		
When All Channels are in Use (2GS/s)	2ms @ 4M points	200μs @ 400k points
When All Channels are in Use (1GS/s)	2ms @ 2M points	500μs @ 500k points
Band Limitation Filters (Analog)		
100MHz	✓ (DS-563x, DS-565x)	
20MHz	✓	✓
2MHz	✓	✓
200kHz	✓	✓

	DS-5600A Series	DS-5400A Series
Band Limitation Filters (Digital)		
LPF	✓	
HPF	✓	
SMA (Simple Moving Average)	✓	
Waveform Capture Mode		
Peak Detection	✓	✓
High-Resolution	✓	
Averaging (units of exponents of 2)	✓ (2 to 65,536)	✓ (2 to 256)
Trigger Function		
Edge Trigger	✓	✓
Edge ALT	✓	
Edge OR	✓	
Event Trigger (cycle, pulse width, dropout)	✓	✓
TV Trigger	✓	✓
Pattern Trigger (AND, NAND, OR, NOR)	✓	
Serial Trigger (UART, SPI, I ² C)	✓	

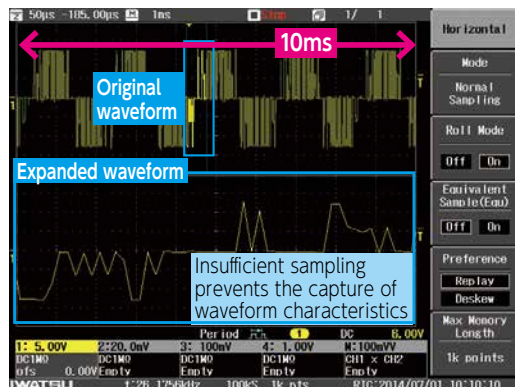
	DS-5600A Series	DS-5400A Series
Waveform Calculation Function		
Addition, Subtraction, Multiplication, FFT	✓	✓
Differential Calculus, Integral Calculus	✓	
Double Calculations (FFT, Differential Calculus, Integral Calculus for Addition, Subtraction and Multiplication)	✓	
Judgment Function		
Waveform Mask Judgment	✓	
Waveform Parameter Judgment (maximum of 4 parameters)	✓	
Waveform Parameter Logging Function		
	✓	
Page Search Function		
	✓	
AUX I/O (only one of each actually mounted)		
CH1/CH2 Output (DS-577)	Factory-delivered Options	
CH1/Trig Output (DS-578)	Factory-delivered Options	
Remote Interface		
LAN	✓	
USB	✓	✓
GPIB	Factory-delivered Options	Factory-delivered Options

Function Introductions

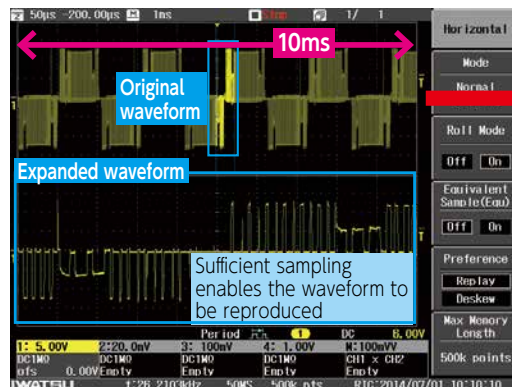
NEW Long Memory up to a Maximum of 5M points DS-5600A Series

[2.5M points/CH when all channels being used] (Maximum of 500k/CH with the DS-5400A Series)

Enables long-term waveforms to be captured while maintaining high-speed sampling.



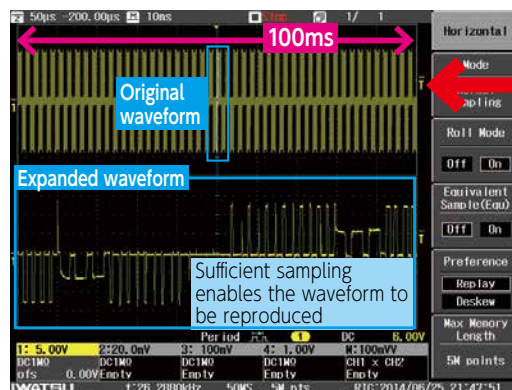
Memory Length: 1k points
Sampling Rate: 100kS/s



Memory Length: 500k points
Sampling Speed: 50MS/s

Waveform Capture Time x 10

The long memory is able to reproduce an even longer waveform capture time to ensure that the entire waveform is acquired so that it can be proportionally checked later.



Memory Length: 5M points
Sampling Speed: 50MS/s

Maximum Sampling Rate for the Waveform Capture Time (DS-5600A Series)

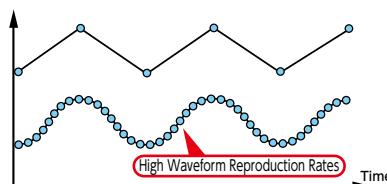
Waveform Capture Time	5M points when the channels are interleaved	2.5M points when all channels are in use
1s	5MS/s	2.5MS/s
100ms	50MS/s	25MS/s
10ms	500MS/s	250MS/s
2ms	2GS/s	1GS/s
1ms	2GS/s	1GS/s

Waveform Capture Time: The s/div x 10div time on the time axis range at the width of the time axis displayed on the oscilloscope.

Long Memory Features

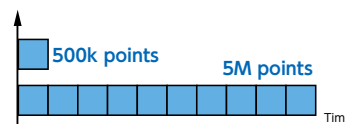
With the same waveform capture time

Waveform reproduction is excellent when expanding a portion of it in the direction of the time axis later by zooming because sampling can be carried out at high speed.



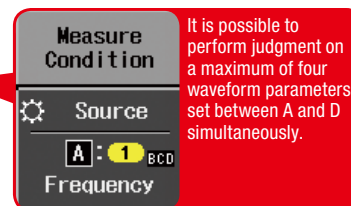
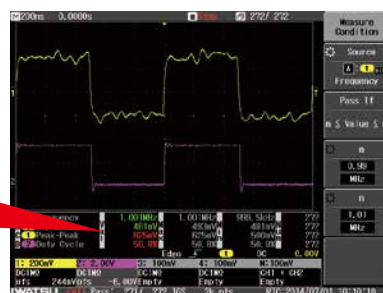
With the same sampling speed

It is possible to lengthen the waveform capture time (time range [s/div] x 10 div)



NEW Four Waveform Parameter Simultaneous Judgment / Waveform Mask Judgment Functions DS-5600A Series

Pass/Fail judgment will be carried out automatically on masks and waveform parameters. Performing this on four parameters simultaneously enables strict conditions to be set.



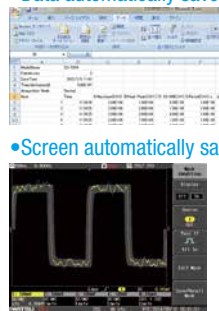
It is possible to perform judgment on a maximum of four waveform parameters set between A and D simultaneously.

Pass/Fail Judgment



Operations during Pass/Fail Judgment

- Waveform capturing halted
- Data automatically saved



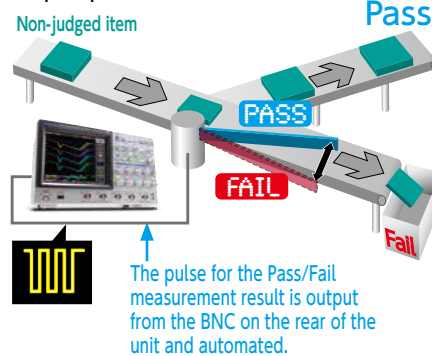
- Pulse output



- Screen automatically saved.
- Beep tone

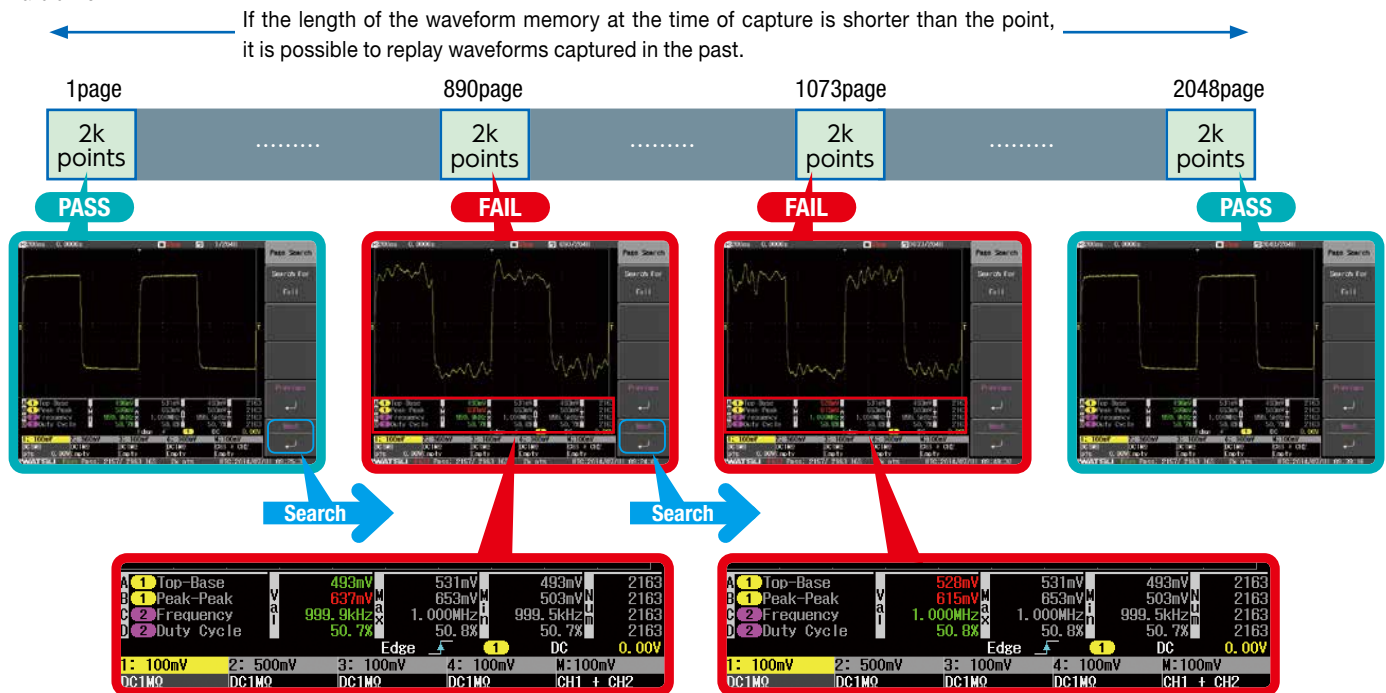


Setup Example:



Page Search Function Using the Judgment Function DS-5600A Series

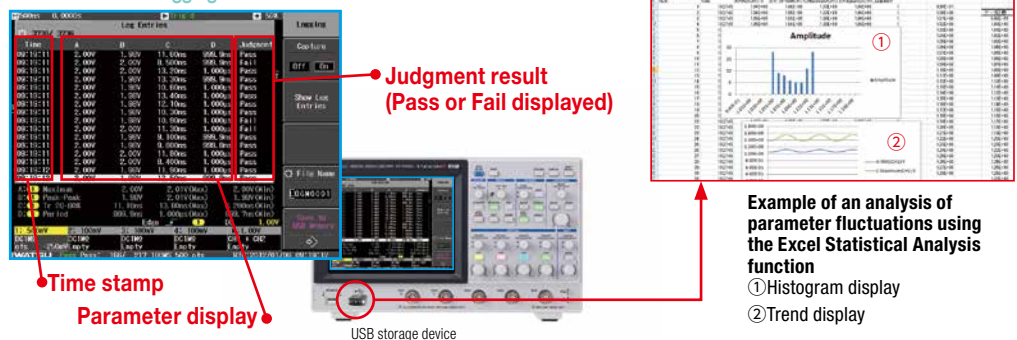
It is possible to search for the results of waveform Pass/Fail results from waveforms automatically saved in the past (a maximum of 2,048 pages with the DS-5600 Series) with the use of the waveform mask judgment and waveform parameter judgment functions, and then display the matching waveforms. This enables us to search waveforms.



Logging Function by Automatic Waveform Parameter Measurement (Displayed in List) DS-5600A Series

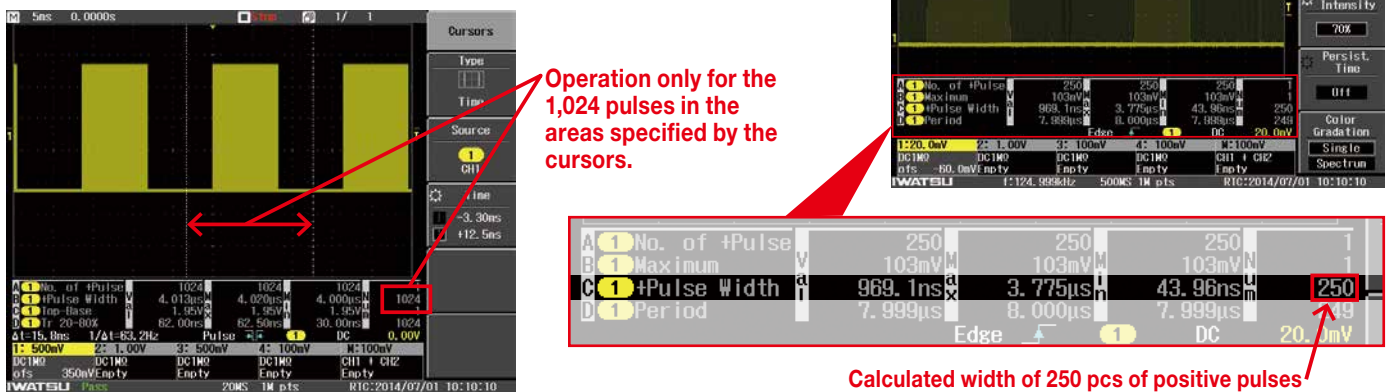
Up to four random waveform parameters can be logged. The logged data can then be saved on a USB memory. Loading this data onto Excel or similar spreadsheet software allows Histograms (discrepancy distribution) and trends (time-based fluctuations) of the waveform parameters to be displayed.

Example of the Automatic Waveform Parameter Measurement at Logging Function



Parameter Auto-measurement for All Cycles* DS-5600A Series DS-5400A Series

Automatically measures the waveforms for all captured cycles (time axis). Loading large amounts of data into the long memory enables the maximum and minimum amplitudes to be isolated. It is also possible to analyze the areas highlighted with the cursors.

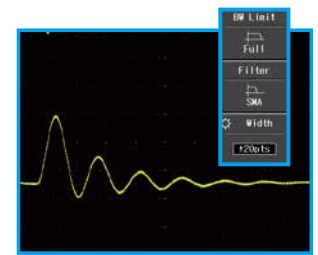
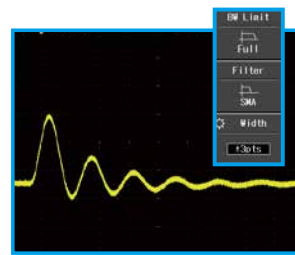
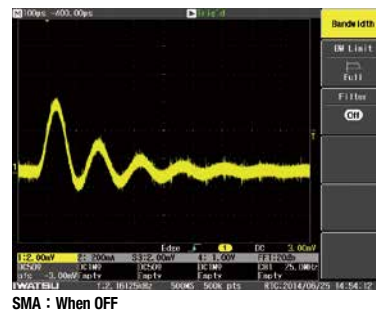


NEW Reinforced Noise Reduction Functions DS-5600A Series

Simple Moving Average

The Simple Moving Average (SMA) enables smoothing and noise reduction at the sampling points of the specified width, through the digital filters that can be set for each channel.

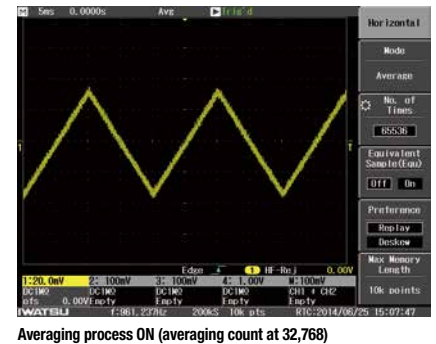
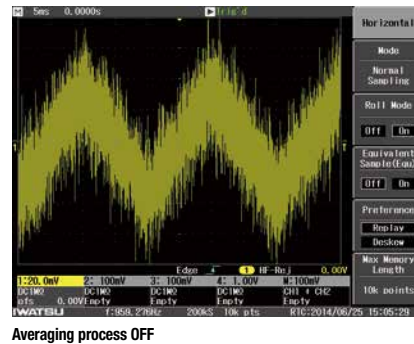
This can also be used on non-repetitive single signals.



Averaging Count Increased

The averaging count setting has been increased from 256 times to 65,536 times. This enables non-synchronized random noise signals to be effectively reduced from measured repetitive signals.

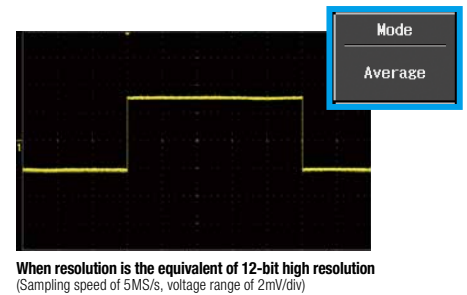
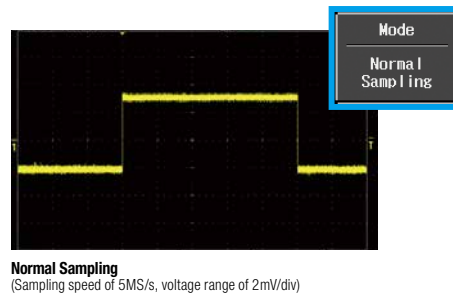
- When the amplitude ratio for the signal (triangular wave) and noise (random) is 1:1
- The example of the right shows a measurement with the sampling speed set at 200k/s and the memory length set at 10k points.



High Resolution

When measurements are taken at a sampling speed lower than the maximum sampling speed, it is possible to average the data captured at the maximum sampling speed, capture the waveforms, reduce random noise, and increase vertical resolution to a level equivalent to a maximum of 12 bits.

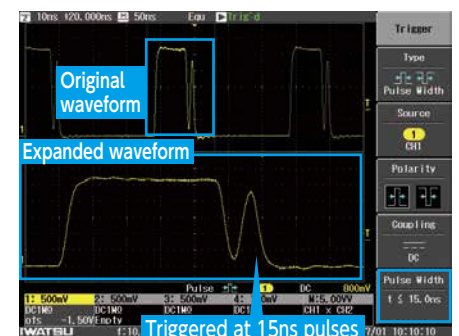
This can also be used on non-repetitive single signals.



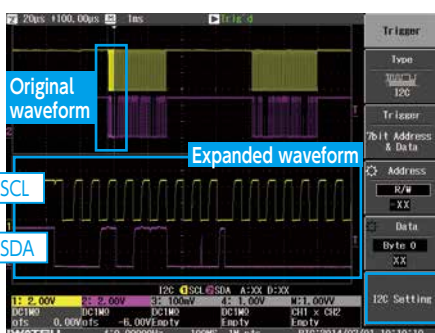
Improved Trigger Functions DS-5600A Series DS-5400A Series

The trigger function has been reinforced so that waveforms can be triggered with optimal conditions, even for complex logic signals and serial data signals. Complex settings performed with pattern triggers can be smoothly set with the use of touch screen operations.

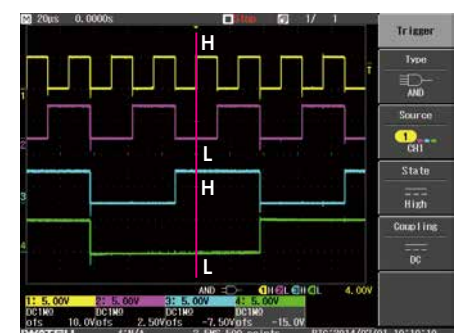
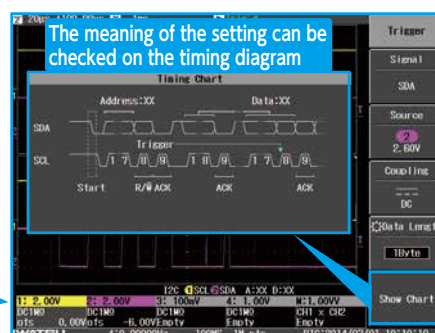
Trigger Types	DS-5600A	DS-5400A
Edge ALT, Edge OR	✓	
Cycle, Pulse width, Dropout, Edge, Pulse count, TV	✓	✓
Pattern	✓	
Serial (UART, SPI, I ² C)	✓	



Pulse Width Trigger
(Example: Detecting abnormal waveforms caused by glitches, etc.)



Serial Trigger
(Example: Observing I²C signals on the serial control bus)

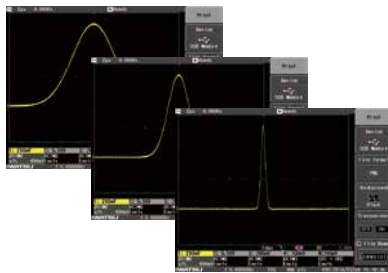


Pattern Trigger
(Example: Counter logic output signal)

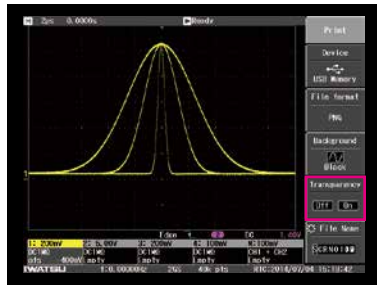
NEW Hard Copy Image Format Reinforced

DS-5600A Series

It is now possible to save images that support transparency attributes when selecting the PNG format. As the background of the waveforms becomes transparent, it is possible to overlay them on documents with PCs for comparison purposes without the use of special image processing.



Normal screen hard copy image (black selected for the waveform background)



Example of screen hard copy images (three waveforms) with black backgrounds overlaid (transparency validated and the images saved)

Rescale Function

DS-5600A Series DS-5400A Series

It is possible to measure values using current probes, shunt resistance and all types of sensors, etc., and unit-convert each of the output voltage signals acquired so that they can be read directly.

Unit Conversion Equation

$$a \times \text{input voltage} + b$$

Vertical axis range zoom rate

Offset

Example during current conversion



The unit can also be selected (V, A, W, °C, none)

Automatic Waveform Parameter Measurement (Horizontal Axis Parameters Measured for All Cycles)

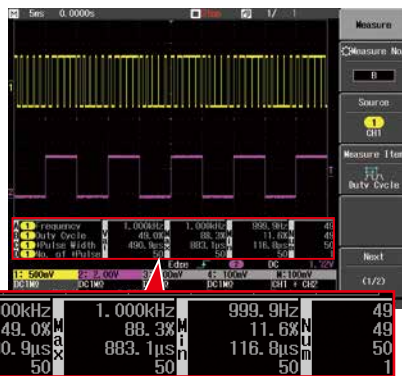
DS-5600A Series DS-5400A Series

It is possible to select and display a maximum of four types of waveform parameters from the twenty-six types available (see P10 and P11 of the specifications), in addition to display of maximum values, minimum values and the measurement count. This function also supports automatic measurement within a range specified by the cursor function. Even further wide-ranging analyses can be performed with the use of the judgment function and the logging function*. * Only supported by the DS-5600 Series

PWM Modulated Waveforms

Carrier frequency: 1kHz
Modulated frequency: 100Hz

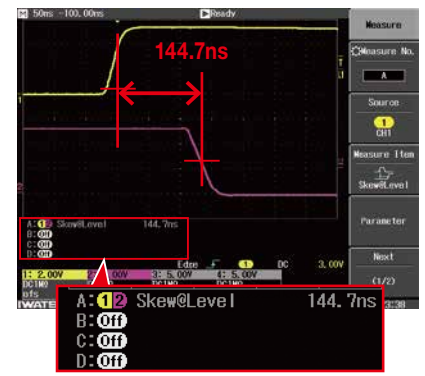
- A: Carrier frequency
- B: Duty ratio
- C: Positive pulse width
- D: Positive pulse count



Skew (Time Difference) Function Between Two Signals

Measures the time difference between the CH1 (rising) and CH2 (falling) logic signals.

The rising/falling parameters and levels for the two signals can be set independently.



Waveform Calculation Function DS-5600A Series DS-5400A Series

Adds, subtracts and multiplies two waveforms, and performs frequency analysis (FFT) on channel waveforms. The DS-5600A Series supports differential and integral calculations.

The calculated waveforms can be saved as data, and can be set as the source for the automatic measurement of waveform parameters.

NEW Supports double calculations (DS-5600A Series)

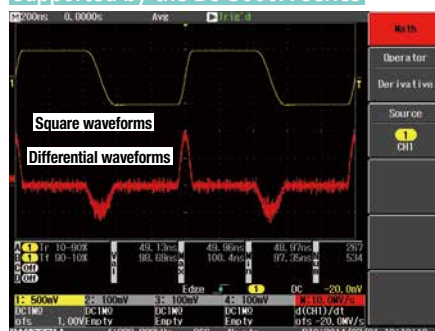
In addition to the results of addition, subtraction and multiplication, this function also supports the double calculation of FFT, differential calculus and integral calculus.

CH Waveforms	Single Operations	Double Operations
1 to 4CH (4CH unit) 1 to 2CH (2CH unit) 2CH among the above	Addition Subtraction Multiplication	FFT Differential calculus Integral calculus
1 to 4CH (4CH unit) 1 to 2CH (2CH unit) 1CH among the above	FFT Differential calculus Integral calculus	
DS-5600	✓	✓
DS-5400	(Excluding differential calculus and integral calculus)	

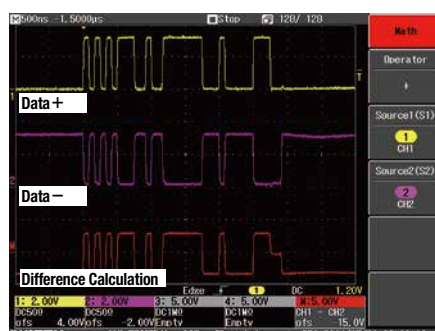
[Examples of Usage]

- Addition/Subtraction: Evaluation of differential signals
- Multiplication: Evaluation of power waveforms from Voltage x Current
- FFT: Analysis of cyclic noise and vibrations, etc., in frequency domains

Supported by the DS-5600A Series

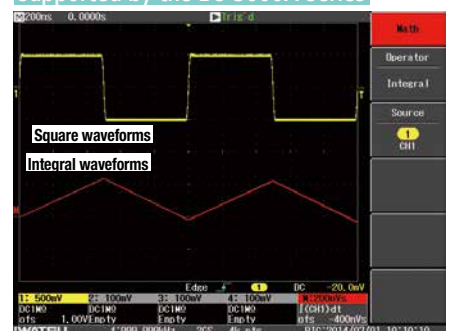


Differential calculation waveforms for square waveforms (rising 50ns, falling 100ns)
(Displays the size of the time fluctuations (dv/dt) for square waveform edges.)



Measuring Differential Serial Signals

Supported by the DS-5600A Series



Integral calculation waveforms for square waveforms
(Displays the results of integral calculus by time (∫vdt) for the area of square waveforms.)



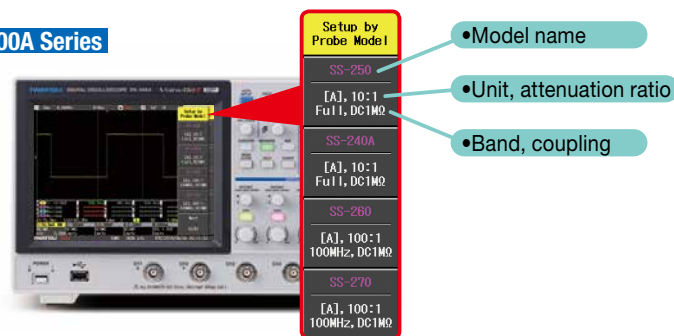
Frequency spectrum analysis (FFT calculations of switching voltage waveforms).

Probe Selection Function DS-5600A Series DS-5400A Series

Selecting probes manufactured by Iwatsu enables attenuation ratios and coupling to be automatically set. The model number, bandwidth of the vertical range and input coupling are displayed.

Eligible Probes

Current Probes:	SS-280 Series, SS-240A, SS-250, SS-260, SS-270
Voltage Probes:	SS-320, SFP-5A, SFP-4A, HV-P30, HV-P60, etc.



Function for Adjusting Skew Between Channels DS-5600A Series DS-5400A Series

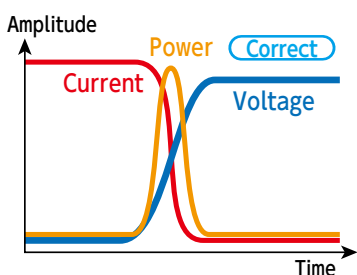
Skew can be adjusted up to a maximum of 500ns*1.

For example, it is possible to obtain even more accurate power waveform calculations*2 by adjusting the phase between the current probe and voltage probe.

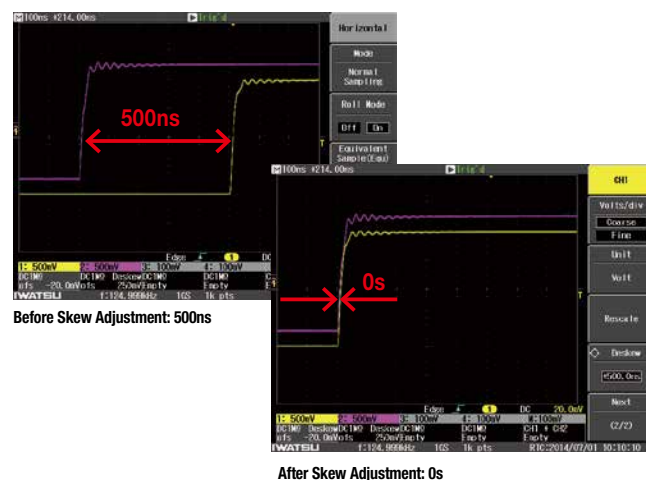
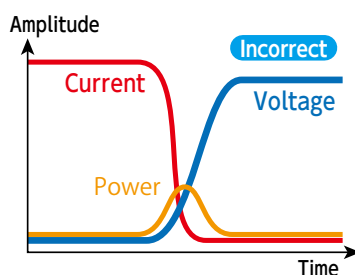
*1 When skew priority is given to the horizontal axis (there are limitations on the number of waveforms that can be used in the replay function).

*2 Difference between power waveform calculations (see the illustration below).

When the Voltage and Current Phases Match



When there is a Large Delay between the Voltage and Current Probes

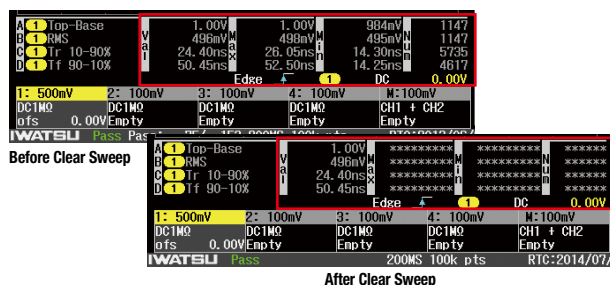


Clear Sweep Function

DS-5600A Series DS-5400A Series

Resets the measurement data for replay waveforms, averaging processes, persistence, waveform parameter auto-measure, Pass/Fail judgment* and logging*, etc.

** Only supported by the DS-5600A Series



High-Speed Remote Waveform Transfer

DS-5600A Series

Provides a remote waveform transfer rate approximately 100times*1 faster than conventional models. This reduces tact time and improves production efficiency when collecting measurement inspection data remotely.

*1: In comparison with the DS-5300 Series using a LAN interface under the waveform display mode set at OFF. (Only supported by the DS-5600A Series)

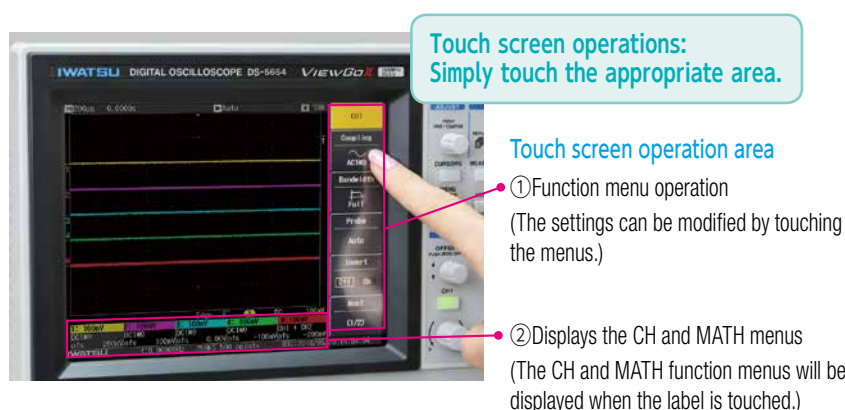
The special conditions represent the measurement results, and the values are for reference purposes.

There are cases in which the actual transmission time will differ depending on the specifications of the PC is use.



Touch screen and independent operation buttons and knobs DS-5600A Series DS-5400A Series

A touch screen function has been mounted on the 7.5-inch color LCD to enable intuitive operations.



Despite its compact size, the independent CH/MATH buttons and knobs enable these functions to be easily used with a single action.



Software Types

Remote Control

Enables vast amounts of data to be collected and high-level analysis to be carried out on PCs.

Scope Viewer (Supplied with Iwatsu Test Instruments Tools)

Download the Iwatsu Test Instruments Tools (free of charge) from the Iwatsu website download page to enable the use of utility software for easily controlling ViewGoII remotely.

Functions: Oscilloscope operations, cursor measurement, waveform data file output, screen hard copies, printing, etc.

Tool Bar
Enables all main operations to be carried out swiftly.

Waveform Display Area
Displays the waveforms and parameter information for each channel. Cursor operations can also be performed with a mouse.

Cursor Measurement Results
Displays measurement results on all channels with the X and Y axes.

Left-hand Pane
Information on the selected window functions are displayed on a tree.

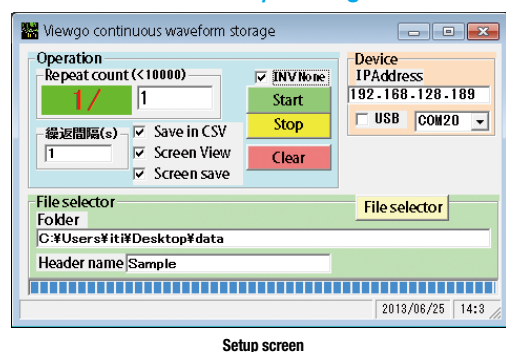
Supported Interface	DS-5600A Series	DS-5400A Series
LAN	✓	
USB	✓	✓

ViewGoII Remote Sample Software

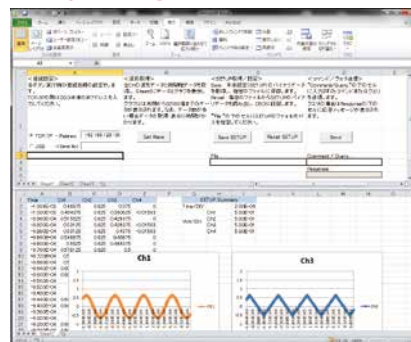
ViewGoII sample software is available on the Iwatsu website download page. Some of the programs are provided in source form so that users can customize them.

Scope Controller (supplied with Iwatsu Test Instruments Tools) is designed to let users make programs with Microsoft Excel, other existing applications, and Microsoft Visual C#, etc., to remote-control ViewGoII via a USB or LAN.

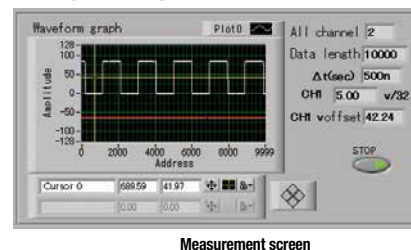
Software for continually loading screen data and CSV data



Example using Excel



Example using LabVIEW

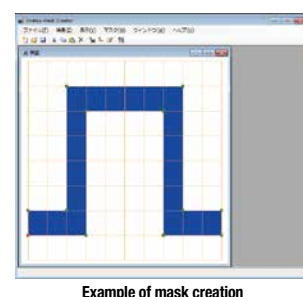


Function: Synchronizes the trigger signals and automatically transfers waveforms to a PC and saves them in files (CSV data, PNG images).

Mask Creation Tool

Mask Creator (supplied with Iwatsu Test Instruments Tools)

An application for creating, editing and saving mask data used for Pass/Fail judgment with ViewGoII (DS-5600A, DS-5600 DS-5500, DS-5500A Series).



DS-5600A Series Specifications



	DS-5654A	DS-5652A	DS-5634A	DS-5632A	DS-5624A	DS-5622A	DS-5614A	DS-5612A
Frequency bandwidth (-3dB)	500MHz		350MHz		200MHz		100MHz	
Rise time (Typical)	750ps		1ns		1.75ns		3.5ns	
Input Channel Count	4	2	4	2	4	2	4	2
Maximum Sampling Rate	2GS/s (when 2 channels interleaved), 1GS/s (when all channels are in use)							
Equivalent Sampling Rate	100GS/s							
Peak detect resolution	1ns							
Averaging	2 to 65536 times (exponent of 2 step)							
Maximum Memory Length	5M points (when 2 channels interleaved), 2.5M points (when all channels are in use)							
Vertical Resolution	8-bit (When high-resolution calculation is valid: Maximum 12-bits)							
Input Voltage Range	2mV/div to 10V/div (1MΩ) , 2mV/div to 2V/div (50Ω)							
Offset Voltage	2mV/div to 50mV/div : ±1V, 50.2mV/div to 500mV/div : ±10V、502mV/div to 10V/div : ±100V							
DC Gain Accuracy	±(1.5% + 0.5% full scale)							
Maximum Input Voltage	±400Vpeak (1MΩ) , 5Vrms (50Ω)							
Band-Limiting Filter	Analog Form: 100MHz, 20MHz, 2MHz, 200kHz Digital Form: Select either LPF, HPF or SMA, 4 independent channels				Analog Form: 20MHz, 2MHz, 200kHz Digital Form: Select either LPF, HPF or SMA, 4 independent channels			
Input Coupling	GND, DC 1MΩ, AC 1MΩ, DC 50Ω							
Input Impedance	1MΩ ±1% // 16pF, 50Ω ±1%							
Probe Sense	Automatic Detection: 1:1, 10:1, 100:1, 1000:1, Manual Settings: 1:1, 5:1, 10:1, 20:1, 50:1, 100:1, 200:1, 500:1, 1000:1, 2000:1							
Time Axis Range	500ps/div to 50s/div		1ns/div to 50s/div		2ns/div to 50s/div		5ns/div to 50s/div	
Standard Probe	SS-101R (multi-channel supplied as standard)				SS-0130R (multi-channel supplied as standard)			
Roll Mode	50ms/div to 50s/div (100kS/s max)							
Clock Accuracy	±10ppm							
Trigger Function	Edge, Edge ALT, Edge OR, Pulse Count, Pulse Width, Cycle, Dropout, TV, Pattern (OR, NOR, AND, NAND), Serial (UART, SPI, I²C)							
TV Trigger (Rated) / Line setting range selection / Field selection	NTSC, PAL, Custom / Up to 3,000 / 1, 2, 4, 8							
Pulse Count Trigger Setting Range	1 to 9,999 events							
Pulse Width Trigger Time Setting Range	15ns to 50s							
Cycle Trigger Time Setting Range	40ns to 50s							
Dropout Trigger Time Setting Range	50ns to 50s							
Pattern Trigger	OR, NOR, AND, NAND							
Trigger Source / State / Threshold Level	All Channels / HIGH, LOW, Don't Care / All Channel Independent Setting							
Serial Trigger								
UART SPI * CH1 input is reserved for SCK signal input: Maximum 20MHz I²C	Trigger Selection	START, STOP, Parity Error, Data Pattern						
	Bit Rate	1,000bps to 1Mbps (set in units of 100bps)						
	Comparative Data Length	5 to 8 bits						
	Signal Source	CH1 to CH4, EXT (CH1, CH2, EXT for 2 channel function)						
	Trigger Selection	Data Pattern						
	CS Selection	Idling time specified when no positive logic/negative logic or CS						
	Comparative Data Length	4 to 64 bits						
	Signal Source	CH1 to CH4, EXT (CH1, CH2, EXT for 2 channel function)						
	Trigger Selection	START, STOP, RESTART, NACK, Data Pattern						
	Address Mode	Selected from 7-bit / 10-bit / EEPROM read						
	Comparative Data Length	1 to 5bytes when the address is 7-bit/10-bit, 1byte when EEPROM read (with shift comparison)						
	Signal Source	CH1 to CH4, EXT (CH1, CH2, EXT for 2 channel function)						
Trigger Source	All channels, EXT (±0.5V), EXT10 (±5.0V), Line							
Trigger Slope / Coupling	+, - / AC, DC, High Frequency Rejection, Low Frequency Rejection, Noise Rejection							
Display / Resolution	7.5-inch Color TFT LCD (touch screen) / VGA: 640×480 Pixels							
Display Mode	Y-T, XY, XY Trigger							
Vector Connection	Sample Point Interpolation Display, Dot Display							
Analog Persistence	Monochrome Grayscale Display, Spectrum Display							
Persistence Display Time	100ms, 200ms, 500ms, 1s, 2s, 5s, 10s, infinite							
Internal Waveform Storage (REF Memory)	5 Waveforms							
Front Panel Setting Storage	Possible to save five settings in the internal memory, USB memory							
Parameter Measurement, Cursor, Zoom, Calculation, Replay Functions								
Parameter Measurement	Maximum Value, Minimum Value, Peak-Peak, RMS, Cycle RMS, Average, Cycle Average, Top, Base, Top-Base, Rising Overshoot, Falling Overshoot, Rising Time 20-80%, Falling Time 80-20%, Rising Time 10-90%, Falling Time 90-10%, Frequency, Cycle, + Pulse Count, - Pulse Count, + Pulse Width, - Pulse Width, Duty Ratio, Integral, Skew (+, -), Skew at level							
Simultaneous Measurement Count / Statistic Value Display	Maximum 4 Parameters / Maximum Value, Minimum Value, Measurement Count							
Logging Items, Output Destination	Time, Parameter Measurement Results (Conditions A, B, C, D), Pass/Fail Judgment Results Recording Time: Pop-up menu, internal memory (maximum 86,400 records), After Recording: USB memory							
Pass/Fail Judgment	Judgment Mode: Parameter Judgment or Mask Judgment, Judgment Results: Saved on USB, Beep Tone, Pulse Output (DS-578 option required), Logging Page Search Function: Select Pass or Fail and search in ascent or descent							
Cursor	Time, Amplitude, Time & Amplitude, Value at Cursor Position							
Zoom	Press the Zoom button on the front panel to display an enlarged waveform on a new grid							
Calculation Function	Addition, Subtraction, Multiplication, Differential Calculus, Integral Calculus, FFT (maximum 8k points, rectangular, hanning, flat-top window functions) Double calculation of the results of either addition, subtraction or multiplication possible with either differential calculus, integral calculus or FFT (9 patterns)							
Rescale / Unit Conversion	A: x + b (x: Input voltage, a, b: User defined) / volt, ampere, watt, °C, no display							
Replay	Automatic waveform logging, storage for a maximum of 2,048 waveforms, replay possible							
Frequency Counter	6 characters							
Interface	Supports USB 2.0HS (device, host), LAN (100Base-TX), GPIB (factory-delivered option DS576), TRIG Output (BNC)							
AUX Interface	Optional external connector							



	DS-5654A	DS-5652A	DS-5634A	DS-5632A	DS-5624A	DS-5622A	DS-5614A	DS-5612A
Optional Accessories								
DS-577 AUX IO CH1/CH2 Output* (factory-delivered option)	AUX IO1: Outputs the CH1 input signal to which offset voltage has been applied, AUX IO2: Outputs the CH2 input signal to which offset voltage has been applied							
DS-578 AUX IO CH1 Output* (factory-delivered option)	AUX IO1: Outputs the CH1 input signal to which offset voltage has been applied							
DS-576 GPIB Interface (factory-delivered option)	GPIB : IEEE488.2							
Power source options for the DS-579 probe	Two-way power source for use with Iwatsu active probes							
Waveform Data Storage	Saved on the USB with binary, ASCII, Mathcad, calculation (ASCII), calculation (Mathcad)							
Hard copy Output	TIFF, BMP and PNG (supporting transparency) images saved on the USB or output to printers that support PictBridge®							
Calibration Signal Output	Square Waveform 1kHz, 3Vp-p							
Power Source / Power Consumption	AC100V to 240V(50/60Hz), AC100V to 120V(400Hz) / 95VA(60W)max							
Dimensions / Unit Weight	Approximately 330W x 190H x 124D mm / Approximately 3.7kg							
Guaranteed Performance Temperature	10°C to 35°C							
Operating Temperature / Humidity / Altitude	Temperature 0 to 40° C / Humidity 5% to 80% RH $\leq$ 30° C (no condensation), RH 55% or less at 40° C or less (no condensation) / Altitude 2,000m or less							

DS-5400A Series Specifications

	DS-5424A	DS-5422A	DS-5414A	DS-5412A
Frequency bandwidth (-3dB)	200MHz		100MHz	
Rise time(Typical)	1.75ns		3.5ns	
Input Channel Count	4	2	4	2
Maximum Sampling Rate	2GS/s (when 2 channels interleaved), 1GS/s (when all channels are in use) 1GS/s			
Equivalent Sampling Rate	100GS/s			
Peak Detect Resolution	1ns			
Averaging Function	2 to 256 times (exponent of 2 step)			
Maximum Memory Length	500k points/ch			
Vertical Resolution	8-bit			
Input Voltage Range	2mV/div to 10V/div			
Offset Voltage	2mV/div to 50mV/div: ±1V, 50.2mV/div to 500mV/div: ±10V, 502mV/div to 10V/div: ±100V			
DC Gain Accuracy	±(1.5% + 0.5% full scale)			
Maximum Input Voltage	±400Vpeak			
Band-Limiting Filter	Analog Form: 20MHz, 2MHz, 200kHz			
Input Coupling	GND, DC 1MΩ, AC 1MΩ			
Input Impedance	1MΩ ±1.0% // 20pF ±2pF (DC1MΩ)			
Probe Sense	Automatic Detection: 1:1, 10:1, 100:1, 1000:1, Manual Settings: 1:1, 5:1, 10:1, 20:1, 50:1, 100:1, 200:1, 500:1, 1000:1, 2000:1			
Time Axis Range	2ns/div to 50s/div		5ns/div to 50s/div	
Standard Probe	SS-0130R (multi-channel supplied as standard)			
Roll Mode	50ms/div to 50s/div (100kS/s max)			
Clock Accuracy	±10ppm			
Trigger Function	Edge, Pulse Count, Pulse Width, Cycle, Dropout, TV			
TV Trigger (Rated) / Line setting range selection / Field selection	NTSC, PAL, Custom / Up to 3,000 / 1, 2, 4, 8			
Pulse Count Trigger Setting Range	1 to 9,999 events			
Pulse Width Trigger Time Setting Range	15ns to 50s			
Cycle Trigger Time Setting Range	40ns to 50s			
Dropout Trigger Time Setting Range	50ns to 50s			
Trigger Source	All channels, EXT (±0.5V), EXT10 (±5.0V), Line			
Trigger Slope / Coupling	+, - / AC, DC, High Frequency Rejection, Low Frequency Rejection, Noise Rejection			
Display / Resolution	7.5-inch Color TFT LCD (touch screen) / VGA: 640×480 Pixels			
Display Mode	Y-T, XY, XY Trigger			
Vector Connection	Sample Point Interpolation Display, Dot Display			
Analog Persistence	Monochrome Grayscale Display, Spectrum Display			
Persistence Display Time	100ms, 200ms, 500ms, 1s, 2s, 5s, 10s, infinite			
Internal Waveform Storage (REF Memory)	5 Waveforms			
Front Panel Setting Storage	Possible to save five settings in the internal memory, USB memory			
AUTO SETUP function	Enabled or disabled			
Parameter Measurement, Cursor, Zoom, Calculation, Replay Functions				
Parameter Measurement	Maximum Value, Minimum Value, Peak-Peak, RMS, Cycle RMS, Average, Cycle Average, Top, Base, Top-Base, Rising Overshoot, Falling Overshoot, Rising Time 20-80%, Falling Time 80-20%, Rising Time 10-90%, Falling Time 90-10%, Frequency, Cycle, + Pulse Count, - Pulse Count, + Pulse Width, - Pulse Width, Duty Ratio, Integral, Skew (+, -), Skew at level			
Simultaneous Measurement Count / Statistic Value Display	Maximum 4 Parameters / Maximum Value, Minimum Value, Measurement Count			
Logging Items, Output Destination	—			
Pass/Fail Judgment	—			
Cursor	Time, Amplitude, Time & Amplitude, Value at Cursor Position			
Zoom	Press the Zoom button on the front panel to display an enlarged waveform on a new grid			
Calculation Function	Addition, Subtraction, Multiplication, FFT (maximum 8k points, rectangular, hanning, flat-top window functions)			
Rescale / Unit Conversion	a * x + b (x: Input voltage, a, b: User defined) / volt, ampere, watt, °C, no display			
Replay	Automatic waveform logging, storage for a maximum of 1,024 waveforms, replay possible			
Frequency Counter	6 characters			
Interface	Supports USB 2.0HS (device, host), GPIB (factory-delivered option DS576)			
AUX Interface	Optional external connector			
Optional Accessories				
DS-577 AUX IO CH1/CH2 Output	—			
DS-578 AUX IO CH1/TRIG Output	—			
DS-576 GPIB Interface	GPIB : IEEE488.2 (factory-delivered option)			
Power source options for the DS-579 probe	Two-way power source for use with Iwatsu active probes			
Waveform Data Storage	Saved on the USB with binary, ASCII, Mathcad, calculation (ASCII), calculation (Mathcad)			
Hard copy Output	TIFF, BMP and PNG (supporting transparency) images saved on the USB or output to printers that support PictBridge®			
Calibration Signal Output	Square Waveform 1kHz, 3Vp-p			
Power Source / Power Consumption	AC100V to 240V(50/60Hz), AC100V to 120V(400Hz) / 95VA(60W)max			
Dimensions / Unit Weight	Approximately 330W x 190H x 124D mm / Approximately 3.7kg			
Guaranteed Performance Temperature	10°C to 35°C			
Operating Temperature / Humidity / Altitude	Temperature 0 to 40° C / Humidity 5% to 80% RH≤30° C (no condensation), RH 55% or less at 40° C (no condensation) / Altitude 2,000m or less			

*The DS-577 and DS-578 cannot be mounted together.

●External appearances and certain performance levels are subject to modification without prior notice for the purpose of product improvement, etc.

Standard DS-5600A Series / DS-5400A Series Prices

Model Name	BW/(-3db)	Channel Count	Maximum Sampling Rate	Memory Length (with CH)	Memory Length (Maximum with 2CH interleaved)
DS-5654A	500MHz	4CH	2GS/s (with 2CH interleaved)	2.5M points	5M points
DS-5652A	500MHz	2CH			
DS-5634A	350MHz	4CH			
DS-5632A	350MHz	2CH			
DS-5624A	200MHz	4CH			
DS-5622A	200MHz	2CH			
DS-5614A	100MHz	4CH	2GS/s (with 2CH interleaved)	500k points	500k points
DS-5612A	100MHz	2CH			
DS-5424A	200MHz	4CH			
DS-5422A	200MHz	2CH			
DS-5414A	100MHz	4CH	2GS/s (with 2CH interleaved)	500k points	500k points
DS-5412A	100MHz	2CH			

* We accept requests for calibration certificates, traceability network diagrams and inspection results on a chargeable basis.

Optional Accessories

• GPIB Interface

DS-576

• AUX I/O Option (2-way probe power source option)

DS-579

Can be used as a power source for probes
Supported Models: SS-240A/SS-250/SS-260/SS-270/SS-320/
SFP-5A/SFP-4A

• AUX I/O Option (CH1/CH2 Output)

DS-577

• AUX I/O Option (CH1 Output)

DS-578

* DS-5600A Series only
* The DS-577 and DS-578 cannot be mounted together.

Rear of DS-5654A

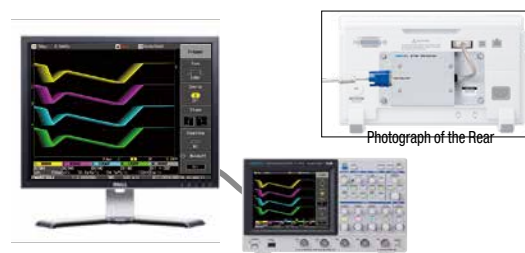
* DS-576, 577, 578 and IE-1226 are factory-delivered options, so it is necessary to specify them when place your order.

VGA Video OUT

IE-1226 Made to order

VGA output on external displays for ViewGo II is possible. In the inspection lines of factories, the efficiency will be improved and in schools, the image onto a large projector screen can be shown.

* The DS-579 cannot be used after the IE-1226 has been mounted.



Photograph of the Rear

Recommended for ViewGo II

Carrying Bag

Models Supported

•DS-5600ASeries, •DS-5600Series,
•DS-5500ASeries, •DS-5500Series,
•DS-5400Series, DS-5400Series



Probe Accessories

*The specifications here show the individual characteristics of each probe.(Contact our sales or distributor for details.)

Standard Probe

SS-0130R

Frequency BW: DC to 200MHz
Input RC: 10MΩ//12.5pF
Attenuation Ratio: 10:1
Length: 1.5m

SS-101R

Frequency BW: DC to 500MHz
Input RC: 10MΩ//12pF
Attenuation Ratio: 10:1
Length: 1.2m

High-Voltage Probe

SS-0170R

Frequency BW: DC to 400MHz
Maximum Input Voltage: 6kV (DC+ACpk, CAT I)
Input RC: 66.7MΩ ± 1%/4pF or less
Attenuation Ratio: 100:1,
Cable Length: 2m

SS-0171R

Frequency BW: DC to 400MHz
Maximum Input Voltage: 4kV(DC+ACpk, CAT I)
Input RC: 66.7MΩ ± 1%/4pF or less
Attenuation Ratio: 100:1,
Cable Length: 2m

High-Voltage Probe

HV-P30A

30kV DC+AC peak or single-pulse 40kV

HV-P60

60kV DC+AC peak or single-pulse 80kV

* Check the de-rating characteristics of the high-voltage probes before selecting them.

High-Voltage Probe

PHV/PHVS Series

Type	BW	Length	Attenuation Ratio	Maximum Input Voltage	
				AC rms(CAT I)	Impulse peak
PHV1000-RO	400MHz	2m	100:1	1kV	4kV
PHVS1000-RO	400MHz	2m	1000:1	1kV	6kV
PHV-641-LRO	380MHz	1.2m	100:1	2kV	4kV
PHV-642-LRO	300MHz	2m			
PHV-643-LRO	150MHz	3m			
PHV661-LRO	380MHz	1.2m	100:1	2.8kV	6kV
PHV662-LRO	300MHz	2m			
PHV663-LRO	150MHz	3m			
PHVS662-LRO	400MHz	2m	1000:1	2.8kV	6kV
PHVS663-LRO	250MHz	3m			

* Contact us with regard to specifications not listed

FET Probe

Model	Attenuation	Input RC	Bandwidth
SFP-5A	10:1	Approx. 1.9pF, Approx. 1MΩ	DC to 1GHz
SFP-4A	10:1	Approx. 2.15pF, Approx. 1MΩ	DC to 800MHz
PS-25	Power supply for SFP-4A, SFP-5A and SS-320 (Input voltage AC100V only)		

SFP-5A



PS-25



High-Voltage Differential Probe

SS-320

DC~100MHz (1kVrms)



Current probe (Clamp type)

SS-250

Frequency Bandwidth : DC to 100MHz(-3dB), Maximum input range : 30A rms,
Maximum peak current : 50A peak, Measurable wire diameter : φ 5mm

SS-240A

Frequency Bandwidth : DC to 50MHz(-3dB), Maximum input range : 30A rms,
Maximum peak current : 50A peak, Measurable wire diameter : φ 5mm

SS-270

Frequency Bandwidth : DC to 2MHz(-3dB), Maximum input range : 500A rms,
Maximum peak current : 700A peak, Measurable wire diameter : φ 20mm

SS-260

Frequency Bandwidth : DC to 10MHz(-3dB), Maximum input range : 150A rms,
Maximum peak current : 300A peak, Measurable wire diameter : φ 20mm

PS-26 Power Source for Current Probes

Power supply for SS-240A, SS-250, SS-260 and SS-270(Input voltage AC100V(AC120V/AC200V/AC220V are factory- delivered options.)

Rogowski Coil Current Probe

SS-280A Series



Model	BW/(-3dB)	Maximum current
SS-281A	110Hz to 30MHz	30A, peak
SS-282A	65Hz to 30MHz	60A, peak
SS-283A	32Hz to 30MHz	120A, peak
SS-284A	9Hz to 30MHz	300A, peak
SS-285A	6Hz to 30MHz	600A, peak
SS-286A	3Hz to 30MHz	1200A, peak

Common to all SS-280A series

Item	Specifications
Cable length	1.5m
Sensor Coil length	80mm
Sensor Coil wire diameter	φ 1.7mm
Temperature range	
Amplifier	0deg. to 40deg.
Coil&cable	-40deg. to 125deg.
Output	BNC connector
Power supply	AA battery *4pcs. or AC adaptor

ex. probe on T0-220 package

•Some of the products are Regulated Products subject to the Foreign Exchange and Foreign Trade Control Law of Japan. Export should not be allowed without appropriate governmental authorization. Please ask our sales office whether the product concerned is a Regulated Product.

•The products shown in this catalogue are current models at the date of publication. Designs and specifications are subject to change without notice for improvement.

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